

Computational Neuroscience Report

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Introduction

In this report we will examine spike train statistics and relations between the Poisson distribution and artificial neuroscience. The Poisson process produces a series of time events, the probability to observe number of events is very similarly shaped to the gaussian bell curve that presents a normal distribution, except the Poisson distribution is a little bit skewed to the side (fig. 1). In the Poisson process, events (often called arrivals) occur independently. - the probability of an event in a Poisson process does not depend on other events. depends solely on the a rate function ($\lambda(t)$) multiplied by a small time gap (Δt). A Poisson process with a constant rate function is called homogeneous. The inter-event intervals of a homogeneous process are exponentially distributed according to the following distribution:

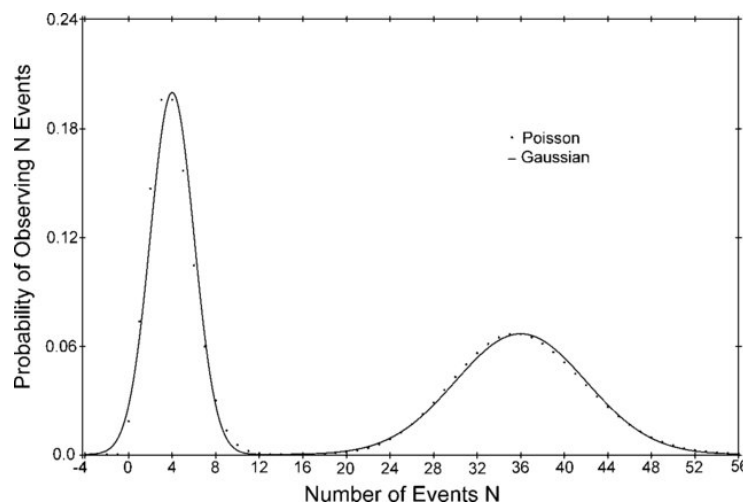
$$p(t) = \lambda e^{-\lambda t}$$

$p(t)$ = probability of event in time t

λ = rate

A process with a varying rate is called inhomogeneous. In the context of neuroscience – action potentials (often regarded as spikes) can be regarded as Poisson events. An action potential is a pulse of voltage generated when a neuron's membrane reaches a certain voltage threshold. Spike trains produced by neurons are considered an inhomogeneous Poisson process because every spike is followed by duration called the refractory period, that (A 2-millisecond-long period on average) that takes place after a spike. During this period, the neuron can't produce a spike. The introduction of the refractory period makes.

Interferes with the constant rate of the homogeneous process – and transforms it into an inhomogeneous process.



The difference between a gaussian distribution(left) and a Poisson distribution(right)

Adapted from: Mark S Rzeszutarski et.al. 1999 (DOI: [10.1148/radiographics.19.3.g99ma33765](https://doi.org/10.1148/radiographics.19.3.g99ma33765))

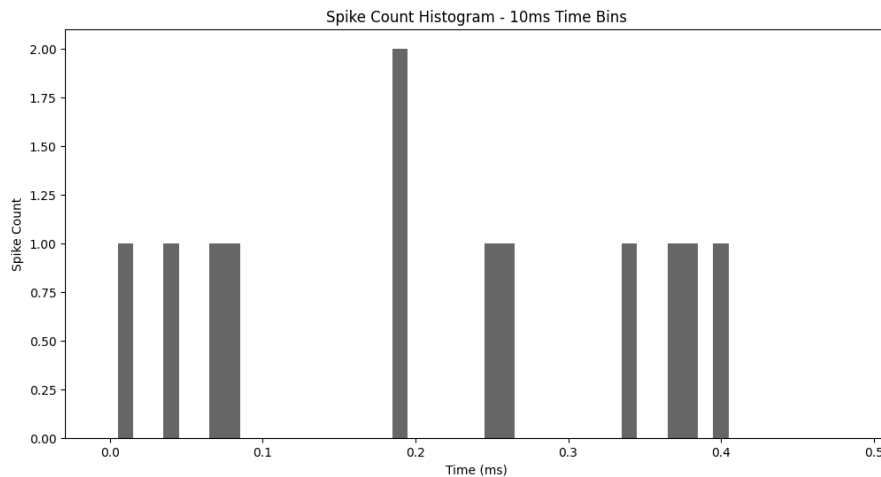
Homogenous Poisson Spike Generator Model:

In our first model, we are simulating a homogeneous Poisson process-based spike-train generator with a fixed fire rate of 35Hz. To do so we sample the Inter-spike-interval (ISI) assuring that its exponential following Equation 1:

```
#generate next spike time using poisson process:
spike_interval = np.random.exponential(1 / fire_rate) #isi
current_time += spike_interval
```

Code snippet demonstrating a skip forward in time by a single ISI to mark the following spike.

The fixed fire rate allows for a constant probability of spike occurrences over time.



The spike count shown above represents the count of spikes per 10ms time-bin.

Its clear that there still is a certain amount of randomness – due to the exponential distribution of the ISIs – exact times between consecutive spike can vary to a degree.

Inhomogeneous Poisson Model with Refractory Period

Building on the initial model, while maintaining the same fire rate of 35Hz, we introduce an absolute refractory period of 5ms to create an inhomogeneous Poisson neuron. This modification prevents spikes from occurring 5ms after a spike has taken place.

To evaluate both models we compared two key metrics, both are 1 for homogeneous processes.

Metric	Formula	Description
Fano Factor	$F = \frac{\text{variance of binned counts } K}{\text{mean of binned counts } K}$	Measures variability of spikes count over time.
Coefficient of variation (CV)	$Cv = \frac{\text{standard of deviation of ISI}}{\text{mean time of ISI}}$	Measures variation of the ISIs.

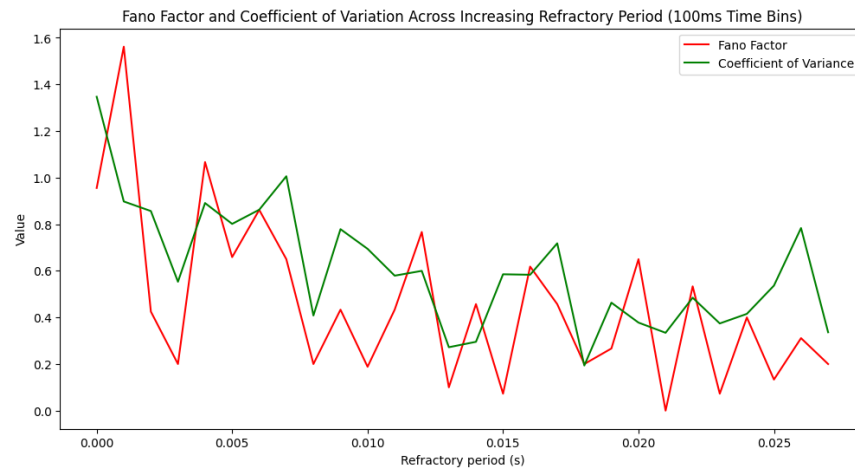
Fano factor:

The Fano factor indicates a neuron's spiking behavior. The behavior is regular for 1, when the Fano is below one – the behavior is more regular than expected for a homogeneous process – as we showed above, a homogeneous process has an amount of randomization. and irregular for $\text{fano} > 1$.

```
fano = np.var(chunk) / np.mean(chunk) #variance/avg spike count per chunk
cvs = np.std(chunked_isi_dur) / np.mean(chunked_isi_dur) #std/avg isi
duration per chunk
```

demonstration of calculation of Fano factor and CV per “chunk”

When introduced to an increasing refractory period (0 – 28 ms), both the Fano factor and CV's values gradually. Decreased. According to the plot – both variability in spike count and ISI decreased the longer the introduced refractory period. An explanation for the observed decrease in Fano factor and CV with increasing refractory period reflects the shift from a more random to a more constrained and regular spiking process. The longer the refractory period, the more deterministic the system becomes, explaining the decline in variability for both metrics.



CV and Fano Factor values plotted against changing refractory period durations spanning 0-28ms.

Leaky Integrate-and-Fire Neuron with Poisson Input

In this section, we simulate a leaky integrate-and-fire (LIF) neuron model.

The leaky integrate and fire neuron model represents a set of complicated biological processes that have been incorporated into code.

$$\tau_m = \frac{dv}{dt} = E_l - v(t) + R_m \cdot i(t)$$

Each variable corresponds to a biological mechanism in the cell.

v = membrane potential (mV)

E_l = reversal potential or resting potential (typically -70mV)

R_m = membrane resistance – the resistance of the membrane (allowed by its highly fatty consistency)

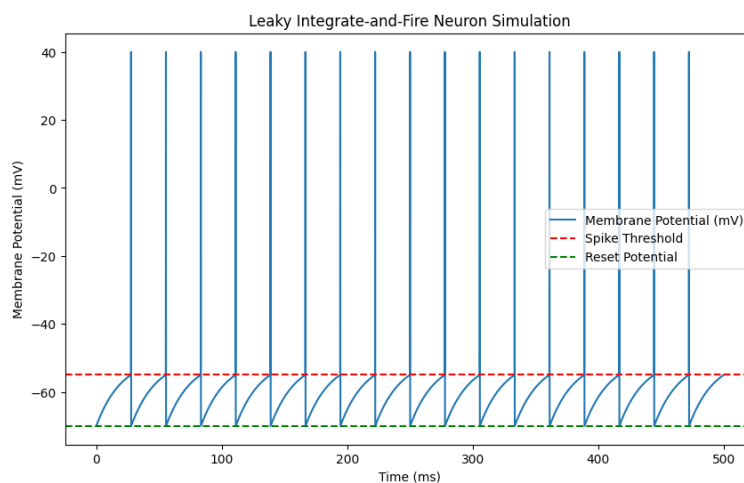
$i(t)$ = often addressed as I_e – current injected into the cell. (Often injected by an electrode.)

```
delta_v = ((E_l - v[t-1]) + (R_m * current_I_e)) / tau_m #lif
v[t] = v[t-1] + delta_v * delta_t #update membrane potential

if v[t] >= v_threshold and v[t-1] < v_threshold: #spikedetection
    spike_times.append(sim_time[t]) #record spike time
    spike_train[t] = 1 #mark spike
    v[t], isSpiking = 40, True #reset after spike
```

code snippet demonstrating usage of the LIF formula to update membrane potential and introduce a condition for spiking in the form of a threshold – often -55mV.

The baseline LIF neuron exhibits deterministic spiking with constant input, resulting in zero variation in ISIs.



Baseline LIF neuron with a constant 2nA current input
Showing a constant ISI

Introducing noisy synaptic input allows us to explore the variability of the LIF neuron's spiking behavior.

To achieve this, we model a poisson generator to distribute a number of inputs symbolizing synapses.

The inputs generated range in values and are plugged into the LIF neuron summed into its input current. To generate random values, we will have to choose the rate values.

To choose the values, we plug a `rate(lambda)` input into the Poisson operator – that indicated for the “strength” of the Poisson distribution. Large rate values would result in more events, or in our case – spikes. For our simulation we will choose most of the values to be weak. The generated values are split into two, one of the arrays will now be turned negative, the addition of a positive and negative synaptic input emulates the activity of inhibitory and excitatory synapses.

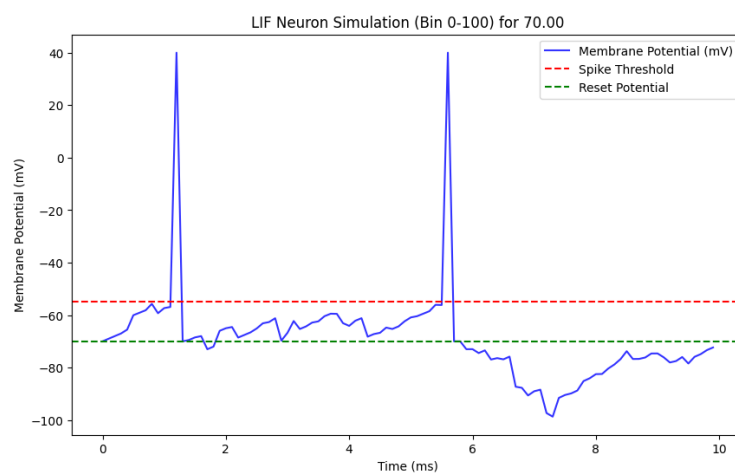
The biological neuron receives signals from its surrounding neurons through synapses in the shape of neurotransmitters – chemicals that trigger a reaction within the neurons membrane some excite it causing its voltage to rise toward the threshold, and some inhibit its voltage accumulation. In our case – excitation values are positive; inhibition values are negative.

```
np.random.shuffle(lambdas)
print("shuffled lambdas", lambdas)
#generate random inhibitory and excitatory weights
inhib_rand = -1 * np.random.poisson(lambdas) #negative inhibitory
excit_rand = np.random.poisson(lambdas)      #positive excitatory
```

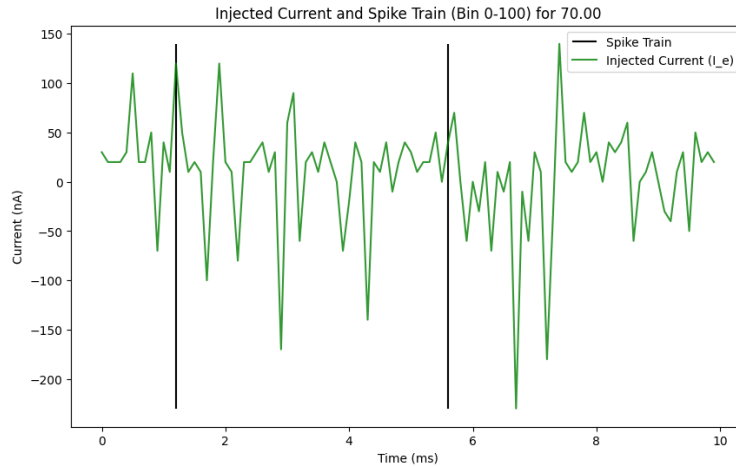
In the snippet we can see the process of Poisson noise generation, the `rate(lambda)` values are being fed into the operator and the operator in turn outputs a Poisson distribution of events. Inhibitory synapse inputs declared as negative.

We analyze the LIF neuron's spiking behavior by computing the **Fano factor** (variance-to-mean ratio of spike counts)

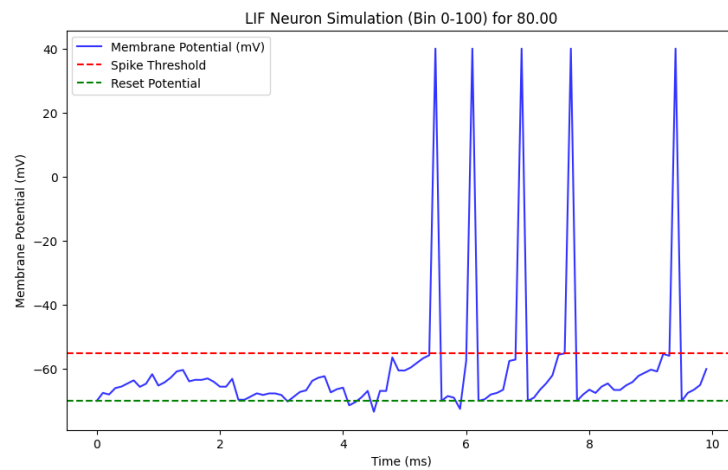
This investigation allows us to assess the impact of stochastic synaptic inputs on the variability of a postsynaptic neuron's spiking behavior, providing insights into how input noise influences neural firing patterns.



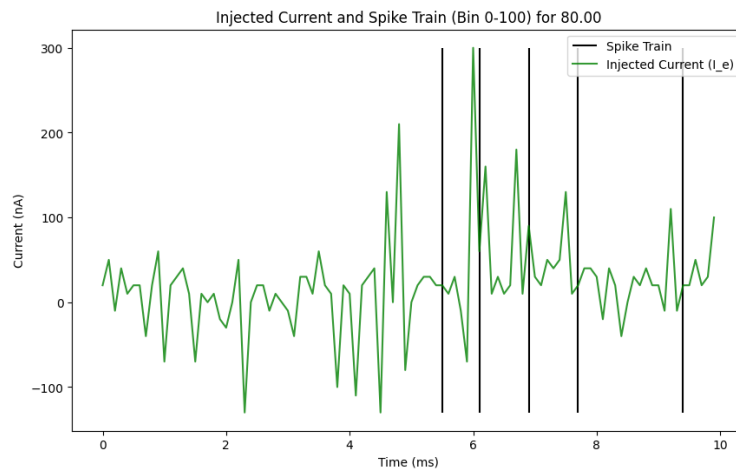
Above is a LIF neuron's membrane voltage response to input current injected with 70% weak Poisson noise and the neuron's spike times.



The fluctuations of an input current injected with 70% weak Poisson noise and the neuron's spike times.



Above is a LIF neuron's membrane voltage response to input current injected with 80% weak Poisson noise and the neuron's spike times.



The fluctuations of an input current injected with 80% weak Poisson noise and the neuron's spike times.

Motion stimulus-evoked spikes

In this analysis we will examine the motion stimulus that evoked the spike train recorded in the file rho.dat. The stimulus data represents the sensory input that provided stimulation for the spiking. To learn more about the relationship between the stimulus and neuronal activity calculate the **spike-triggered average (STA)**, which presents the average stimulus preceding a spikes.

$s(\tau)$: The output, the Spike-Triggered Average (STA).

n : The number of spikes.

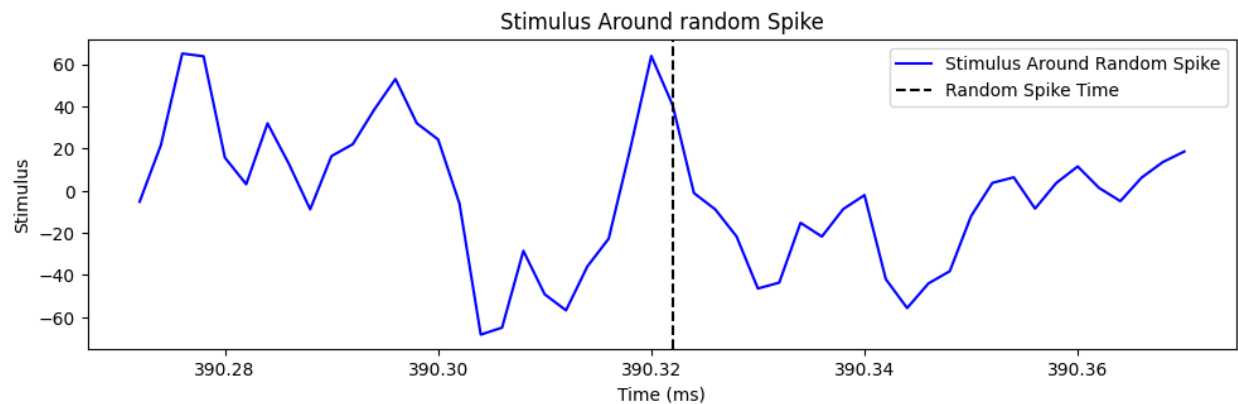
$x(t_i - \tau)$: The signal before each spike.

t_i : The time points at which spikes occur.

τ : The time window size before each spike.

```
#calc STA
#stim corresponds to xi (signal data)
#spike_times corresponds to ti (array of spike times)
#window corresponds to tau (window size)
#len(spike_times) corresponds to ni (number of spikes)
sta = np.zeros(window)
for t in spike_times:
    if t >= window:
        sta += stim[t-window:t]
sta /= len(spike_times)
```

$$s(\tau) = \frac{1}{n} \sum_{i=1}^n x(t_i - \tau)$$



The sensory stimulus preceding and following a spike. Calculated over 100ms window.

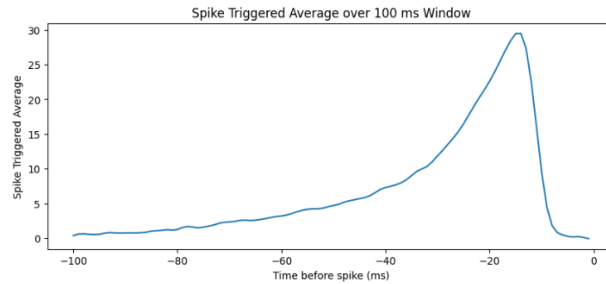
The STA is calculated over a 100ms window, capturing the temporal relationship between sensory stimulus features and neuronal responses.

This process involves aligning the stimulus data to each spike in the rho.dat vector and averaging across all spikes within the defined time window.

The resultant STA plot highlights the underlying cause of the centered spike, shading more light on the processes that preceded a spike

Stimulus Triggered by Pairs of Spikes

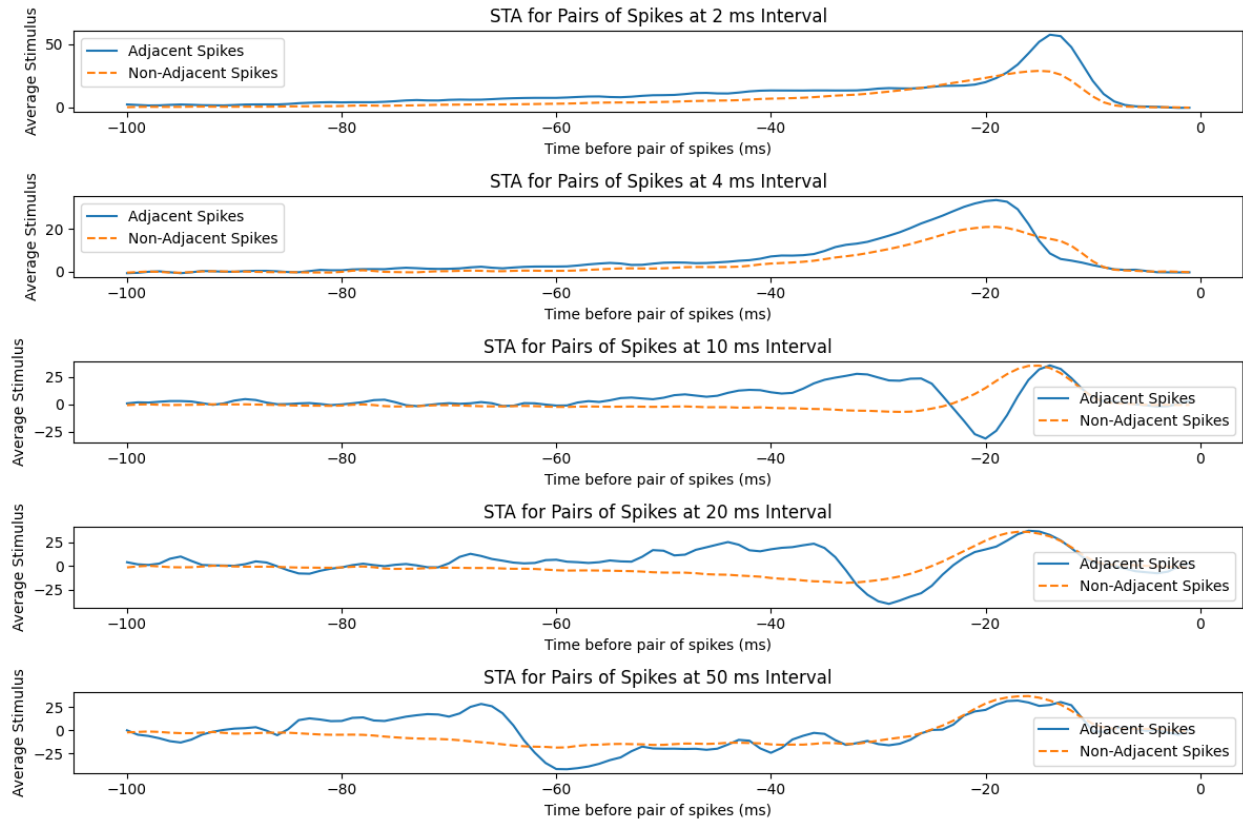
For this part of the report we will explore to what degree does the proximity of spikes relate to the preceding stimulus. We present the data in various interval windows (2 ms, 4 ms, 10 ms, 20 ms, and 50 ms).



STA over a 100ms window - no regard to adjacency.

For this part of the analysis, we explore how the proximity of spikes relates to the preceding stimulus. Using various interval windows (2 ms, 4 ms, 10 ms, 20 ms, and 50 ms) we calculate and plot the **STA** preceding a spike pair. We are addressing two cases where the spikes are either adjacent (directly next to each other) or non-adjacent (separated).

The difference between the two cases might serve as an insight into the relation between the sensory stimulus and the spike.



Plotted above are the differences between adjacent spikes and nonadjacent spikes.

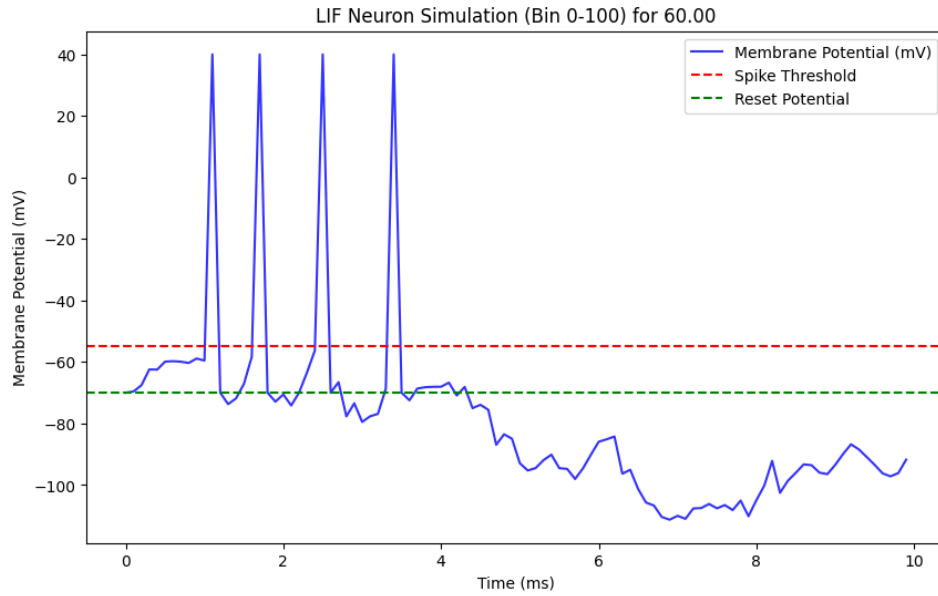
As a general trend we can see that the average stimulus for adjacent spikes starts fluctuating tens of milliseconds before the spike, unlike the nonadjacent stimulus which is mostly smooth and stable.

A hypothesis regarding the addition of a refractory period to an adjacent spike could be great interference with the following spike, and possibly reduction in both the frequency and the amplitude of the spike, where a nonadjacent spike spans shorter than an adjacent spike, and so it would be less affected by a refractory period.

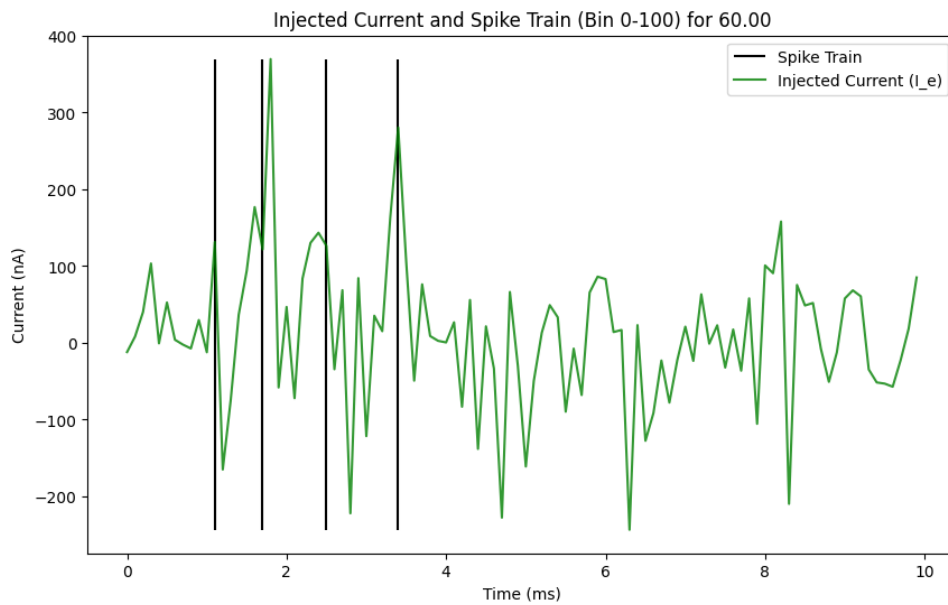
Comparison with Integrate-and-Fire Neuron Model

In this analysis, we extend our investigation by simulating an **integrate-and-fire (LIF) neuron model** and comparing its behavior to the recorded H1 neuron. The LIF neuron receives input based on the stim.dat stimulus, with an additional constant current added to ensure firing. The stimulus' strength can be varied to assess its impact on the neuron's firing dynamics. We explore the effect of varying the strength of the stimulus contribution on the spike train characteristics. This simulation allows us to understand how the LIF neuron processes the stimulus and whether the model's behavior aligns with the recorded experimental data, providing insights into the underlying mechanisms of the biological neuron's response.

The following figures show the response of a neuron (pre injected with stim.dat current input) to an increasing weak - Poisson noise value percentage:

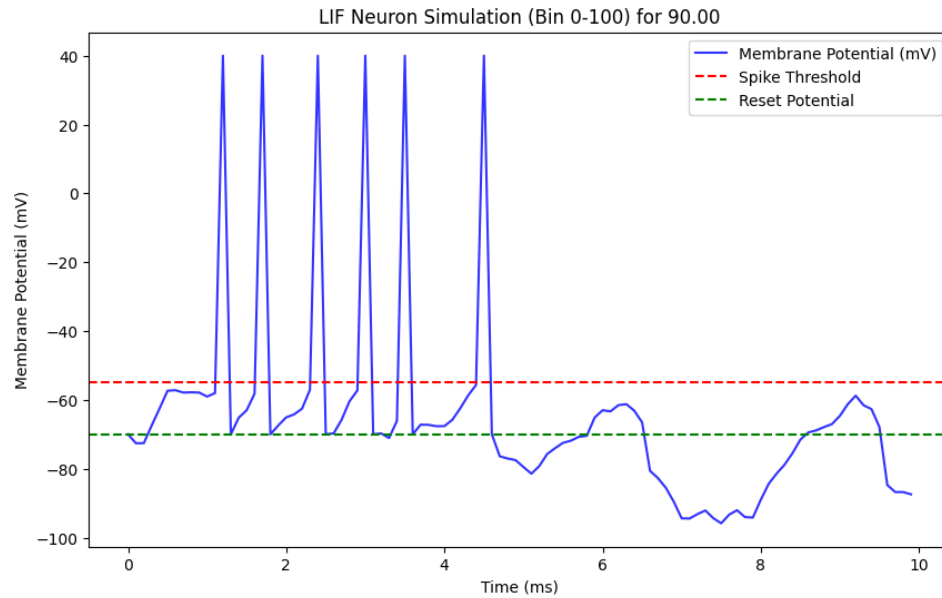


Above is a LIF neuron's membrane voltage response to input current injected with 60% weak Poisson noise and the neuron's spike times.

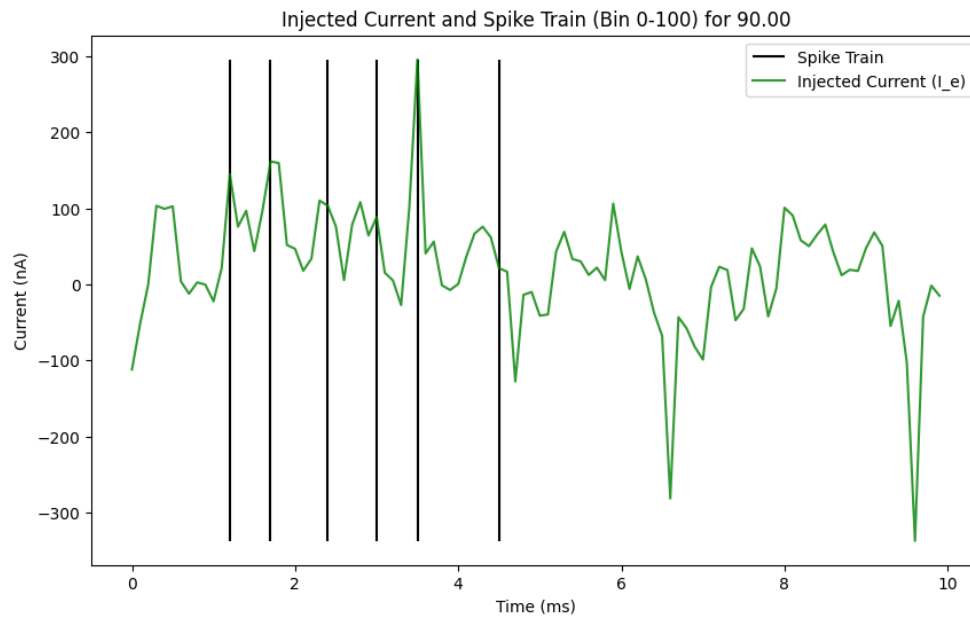


The fluctuations of an input current injected with 60% weak Poisson noise and the neuron's spike times.

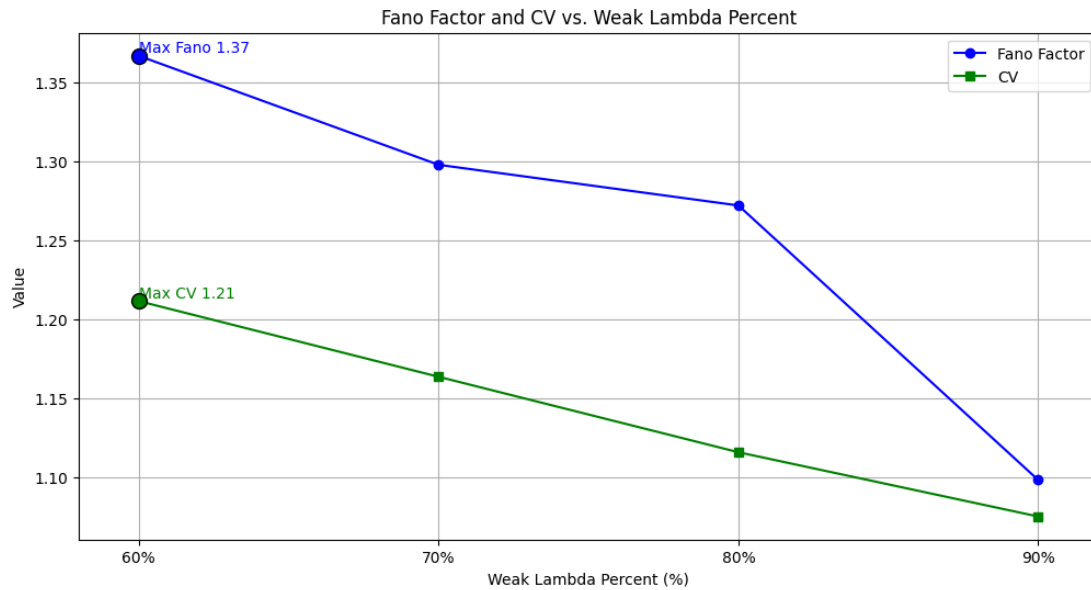
There seems to be a resemblance to the shape of the sensory stimulus nearing the spikes.



The same LIF neuron's membrane voltage response to an input current injected with 90% weak Poisson noise and the neuron's spike times.



The fluctuations of an input current injected with 90% weak poisson noise and the neuron's spike times.



Fano Factor and CV values as a function of the increasing percentage of weak Poisson values: This figure is remarkably interesting because it supports the claim that Fano Factor and CV values decrease upon a shift to a more regular pattern – 90% weak Poisson values introduces a more regular pattern.

References:

Rzeszutarski (1999). *The AAPM/RSNA Physics Tutorial for Residents* 1. *Radiographics* 19(3):765-782. DOI: 10.1148/radiographics.19.3.g99ma33765